

Development of a vibrotactile stimulation system for cognitive rehabilitation

Master Thesis

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Abstract

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1 Introduction

1.1 Motivation

Dementia is one of the greatest medical and social challenges of the 21st century. More than 55 million people are affected worldwide - and the trend is rising [1], [2]. Despite advances in medication, treatment remains limited, particularly in terms of cognitive abilities.

Non-pharmacological interventions are therefore increasingly coming into focus [3]. Of particular interest are sensory-based approaches such as vibrotactile stimulation, which can activate specific areas of the brain via targeted stimuli. Initial animal studies show positive effects of 40 Hz stimulation on the reduction of amyloid deposits and on cognitive performance [4], [5], [6].

1.2 Objectives of the Thesis

The aim of this work is to develop a technical system for implementing such stimulation methods. The focus is on a precise, reproducible actuator system that is suitable for applications in the field of cognitive rehabilitation.

1.3 Structure of the Thesis

This work is theoretically and technically oriented and therefore begins with a chapter on the necessary basics. The aim is to provide the medical and technological background knowledge required to understand the system developed.

2 Theoretical Background

This chapter provides the theoretical foundation for the development of the vibrotactile stimulation system. It is divided into two main parts. The first part introduces the **medical background**, including neurodegenerative diseases such as Alzheimer's and Parkinson's disease, their associated cognitive impairments, and current approaches to cognitive rehabilitation. The second part focuses on the **technical background**, describing the principles of vibrotactile stimulation, actuation technologies, and existing systems. This structure ensures that both the clinical motivation and the technical requirements are clearly outlined before presenting the system analysis and implementation in the following chapters.

2.1 Medical Background

2.1.1 Neurodegenerative Diseases with Cognitive Impairments

Neurodegenerative diseases such as Alzheimer's disease and Parkinson's disease are among the leading causes of cognitive impairment in older adults. They are characterized by a progressive loss of neuronal structures and functions, which over time leads to marked deficits in memory, attention, executive abilities, and the capacity to manage everyday life [7].

Alzheimer's Disease

Alzheimer's disease is the most common form of dementia, accounting for approximately 60 % to 70 % of all cases [2]. According to the Alzheimer's Association, more than 55 million people worldwide are currently affected [1]. The disease is characterized by the pathological accumulation of proteins, including amyloid- β plaques and τ -protein-related neurofibrillary tangles, which lead to the progressive death of neurons [8].

The first noticeable symptoms usually involve a decline in episodic memory, the part of long-term memory responsible for personal experiences. As the disease advances, deficits in language, spatial orientation, and judgment become more pronounced. Alzheimer's disease is long-lasting and progressive, eventually resulting in complete dependence on care in the later stages [2].

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Parkinson's Disease

After discussing Alzheimer's disease, the second major neurodegenerative disorder with cognitive impact is Parkinson's disease. Parkinson's disease is primarily known as a movement disorder, but cognitive symptoms are also frequent. Approximately 30 % to 40 % of patients develop Parkinson's dementia during the course of the disease [9]. Typical impairments include deficits in attention control, executive functions, and visuospatial perception, while memory is often less affected in the early stages [10].

The underlying cause of the disease is a shortage of dopamine in the brain, resulting from the progressive death of neurons in the *substantia nigra*. In addition, the accumulation of pathological protein deposits, known as Lewy bodies, can further contribute to the development of cognitive symptoms [11].

Cognitive Deficits and Functional Impact

In addition to these disease-specific aspects, it is important to consider the resulting cognitive deficits and their functional impact. Cognitive impairments in neurodegenerative diseases affect core functions such as memory, attention, language, visual processing, and executive abilities [7]. These deficits have a direct impact on everyday activities, for example when taking medication independently, managing finances, maintaining social contacts, or navigating in space [12].

In addition to these practical limitations, emotional and psychosocial burdens are common. Patients often experience frustration, depressive moods, and social isolation, which increase the strain on both those affected and their families.

Therefore, the early detection of cognitive symptoms is of great importance. It provides the basis for targeted interventions and rehabilitation measures that aim to preserve independence and quality of life for as long as possible [13].

2.1.2 Cognitive Rehabilitation: Concepts, Methods, and Target Groups

Multidisciplinary approaches are an important part of cognitive rehabilitation [3]. They combine cognitive training, motor therapy, music therapy, and sensory interventions. The goal is to maintain the independence of patients for as long as possible and to reduce the psychosocial burden on both patients and caregivers.

In addition to traditional training methods, EEG-based biomarkers are increasingly used. Indicators such as the Brain Symmetry Index (BSI) and the Laterality Coefficient (LC) allow an objective assessment of the brain's functional state. EEG analysis makes it possible to adapt rehabilitation strategies to the individual by detecting changes in brain activity, especially in connection with *motor imagery*, which is an established rehabilitation method. These approaches are mainly used with stroke patients, who often suffer from both motor and cognitive impairments.

2. Theoretical Background

Another important method within cognitive rehabilitation is sensory stimulation, with a growing focus on low-frequency vibration. Campbell et al. [14] report in their scoping review that vibrotactile interventions can improve attention, mood, and cognitive stability. Depending on the application context, ranging from short-term interventions in nursing homes to long-term home-based programs, studies found positive outcomes in standardized cognitive tests as well as good patient acceptance.

In summary, cognitive rehabilitation is not limited to one single method but is based on a complementary, multimodal approach. The integration of cognitive training, EEG-based biomarkers, and sensory methods such as vibrotactile stimulation forms a broad field of interventions that is likely to become more important in the future.

2.2 Technical Background

2.2.1 Vibrotactile Stimulation: Principles and Therapeutic Applications

Vibrotactile stimulation refers to the application of mechanical vibrations to the skin in order to modulate sensory, motor, or cognitive processes. The fundamental principle is the activation of mechanoreceptors such as Pacinian and Meissner corpuscles, which are particularly sensitive to low- and mid-frequency vibrations [14]. Through afferent pathways, these peripheral inputs can influence central neural networks and activate both cortical and subcortical structures.

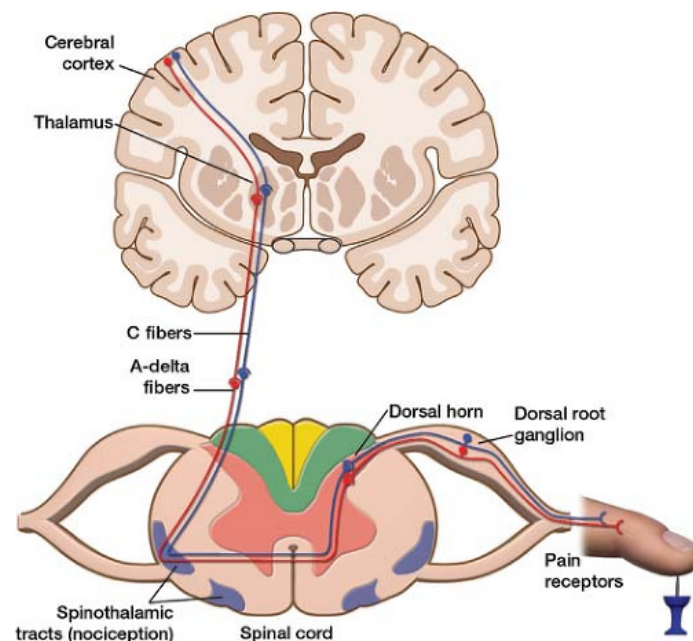


Figure 2.1 PainTransmitted

2. Theoretical Background

The therapeutic use of vibrotactile stimulation has been investigated for several decades. Early studies demonstrated beneficial effects in the rehabilitation of neurological conditions, particularly in patients with motor impairments resulting from stroke or Parkinson's disease [15, 16, 17, 18]. Vibrotactile input has been shown to improve muscle control, reduce tremor, and promote plasticity within the motor cortex.

In addition, vibrotactile stimulation has been applied in the context of music therapy. Clements-Cortés and colleagues have shown that music-vibrotactile interventions (MVT) can enhance quality of life and mood in patients with neurodegenerative diseases [19, 20, 21, 22]. The integration of auditory and somatosensory input strengthens the multisensory nature of therapy and facilitates access to emotional and cognitive processes.

Positive effects have also been reported in dementia and Alzheimer's disease. Vibrotactile stimulation has been associated with reductions in agitation and restlessness, as well as improvements in social engagement and emotional well-being [23, 22]. According to Campbell et al. [14], vibrotactile stimulation may not only provide symptomatic relief but also trigger long-term neuroplastic mechanisms.

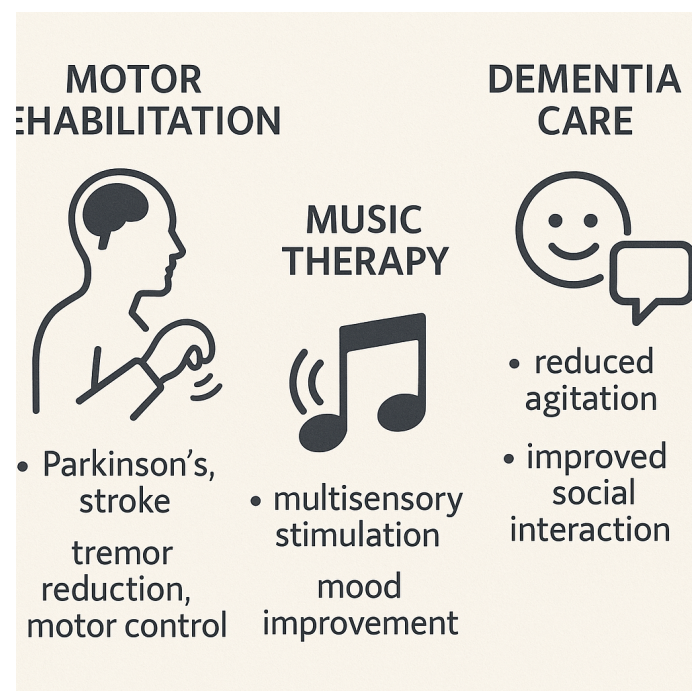


Figure 2.2 Applications Therapy

Overall, vibrotactile stimulation emerges as a promising therapeutic strategy across neurological rehabilitation and music therapy. Its effectiveness relies on the precise activation of peripheral mechanoreceptors and the induction of central nervous system responses, which vary according to frequency, intensity, and duration. This versatility opens opportunities for a wide range of clinical applications, spanning from tremor

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reduction to improvements in emotional well-being and quality of life in patients with cognitive disorders.

2.2.2 Actuation Technologies for Haptic Feedbacks

The development of actuation technologies is central to the implementation of haptic feedback systems. Such systems rely on transducers capable of delivering precise vibrotactile stimuli at defined frequencies and amplitudes, thereby enabling targeted engagement of somatosensory receptors. Commonly used actuators include eccentric rotating mass (ERM) motors, linear resonant actuators (LRA), and piezoelectric transducers, each differing in frequency range, energy efficiency, and capacity for fine-grained control. Piezoelectric actuators, for example, allow for high-frequency and localized stimulation, making them particularly suitable for experimental and therapeutic applications where temporal precision is critical.

Of particular relevance in recent years has been the exploration of vibrotactile stimulation at 40 Hz and within the γ -frequency-band. These frequencies have attracted significant interest due to their documented neurobiological effects. Iaccarino et al. [5] demonstrated that 40 Hz visual stimulation can entrain γ oscillations in the brain and reduce amyloid- β load in Alzheimer's disease mouse models. Extending this line of evidence, Martorell et al. [6] reported that combined auditory and somatosensory 40 Hz stimulation promoted widespread γ -entrainment, improved memory performance, and reduced neuropathological markers in Alzheimer's models. Such findings suggest that haptic actuators capable of delivering stable γ -band vibrations may be of therapeutic relevance in neurodegenerative conditions.

The broader neurobiological importance of γ -oscillations has been highlighted in both animal and human studies. Maby and Colgin [4] reviewed the functional role of γ -rhythms in memory encoding and retrieval, emphasizing their contribution to cognitive integration processes. In a similar vein, Reiko Tutida et al. [24] provided evidence that γ -range stimulation can modulate neuronal synchronization and cortical excitability. These insights reinforce the rationale for designing haptic technologies that can reliably deliver vibrotactile stimulation in the γ -frequency range.

Taken together, advances in actuation technologies not only enable precise delivery of vibrotactile feedback but also open new therapeutic possibilities. By harnessing the neurobiological effects of γ -band stimulation, haptic systems may serve as tools for cognitive enhancement and as non-invasive interventions in neurodegenerative disease management.

2.2.3 Overview of Existing Vibrotactile Stimulation Systems

In recent years, a variety of vibrotactile stimulation systems have been developed for both research and clinical purposes. These systems differ in their actuation technologies,

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form factors, and therapeutic objectives, yet they share the common goal of engaging the somatosensory system through controlled vibratory input.

One prominent class of devices are *vibrotactile chairs* and *sound beds*, which integrate low-frequency transducers into seating or lying platforms. Such systems are widely used in music therapy and dementia care, enabling whole-body vibrotactile stimulation combined with auditory input to promote relaxation, mood improvement, and social engagement [19, 20, 22].

In contrast, *wearable vibrotactile devices* have been designed to provide localized stimulation to specific body regions. These include wristbands, armbands, and vests equipped with small actuators such as linear resonant actuators or piezoelectric elements [17, 18]. Wearable solutions are particularly attractive for mobile health applications, as they enable individualized therapy and integration with portable monitoring technologies such as EEG and biosensors.

A third category comprises *research platforms* that employ arrays of actuators to deliver spatially and temporally precise vibrotactile stimuli. These are frequently used in experimental neuroscience and brain–computer interface (BCI) studies to investigate somatosensory processing, attention, or neuroplasticity [25, 26]. Some systems further combine vibrotactile stimulation with auditory or visual input to exploit multisensory integration mechanisms.

Overall, existing vibrotactile stimulation systems range from stationary therapeutic platforms to mobile wearables and experimental setups. Their diversity reflects both the wide spectrum of application contexts—from relaxation and dementia care to motor rehabilitation and BCI research—and the rapid technological progress that enables increasingly precise and user-centered designs.

2.2.4 Voice Coil Actuators for Vibrotactile Stimulation

Voice coil actuators (VCAs) represent a class of electromagnetic transducers that generate linear motion through the interaction of a coil and a permanent magnet. Unlike eccentric rotating mass (ERM) motors or linear resonant actuators (LRA), VCAs offer high precision, a wide frequency bandwidth, and rapid response times, making them particularly suitable for delivering controlled vibrotactile stimulation. Their ability to generate low-distortion vibrations across a broad frequency spectrum allows for the implementation of experimental protocols that demand both temporal accuracy and physiological relevance.

Recent developments in brain–computer interface (BCI) and wearable electroencephalography (EEG) technologies provide a framework for integrating VCAs into closed-loop neurostimulation systems. Guger et al. demonstrated that portable EEG systems can be effectively applied in real-world contexts, extending the scope of BCI applications beyond laboratory settings [25, 27]. Building on this, Sebastian-Romagosa et al. [26]

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highlighted the potential of mobile EEG in conjunction with tactile stimulation for neurorehabilitation, where wearable systems allow continuous monitoring of brain responses during multimodal interventions.

The integration of VCAs with EEG-based monitoring further supports the development of adaptive stimulation paradigms. Xu et al. [28] reviewed recent advances in wearable EEG systems, emphasizing their growing role in mobile health applications and in the design of neuroadaptive devices. In this context, VCAs provide a versatile platform for vibrotactile actuation, capable of delivering stimulation at therapeutic frequencies (such as the γ -range) while EEG simultaneously captures neural entrainment and cortical responses.

Consequently, voice coil actuators combined with wearable EEG and BCI technologies represent a versatile platform for adaptive vibrotactile stimulation. Their capacity to integrate precise stimulation delivery with simultaneous monitoring of neural responses underscores their potential as a cornerstone of next-generation neurotechnology.

In conclusion, the presented concepts and technologies form the technical basis for the subsequent system development. By examining the principles of vibrotactile stimulation, the range of actuation technologies, existing applications, and the specific advantages of voice coil actuators, this chapter establishes the foundation for the detailed analysis of the current VCA-based setup introduced in Chapter 3.

3 Analysis of the Voice Coil Actuator System

Based on the theoretical background presented in Chapter 2, this chapter focuses on the practical aspects of the voice coil actuator (VCA). First, the general structure and mathematical modeling of VCAs are discussed, followed by a detailed description of the specific system used in this study and its experimental characterization.

3.1 General Structure and Mathematical Modeling of Voice Coil Actuators

A voice coil actuator (VCA) is an electromechanical device that directly converts electrical energy into linear motion [29, 30]. Its structure consists of several essential components that jointly determine its performance.

At the core of the actuator is the **permanent magnet assembly**, typically made from high-energy materials such as NdFeB. Together with a ferromagnetic return path, usually soft iron or steel, the permanent magnet forms a magnetic circuit that channels flux into a narrow air gap. This concentrated flux region ensures efficient electromechanical conversion [31].

Placed within the air gap is the **moving coil**, wound around a lightweight bobbin. In the moving-coil design, current flowing through the winding generates a Lorentz force proportional to the magnetic flux density [32, 30]. Alternative moving-magnet configurations also exist, but they are less common due to increased thermal challenges [29]. The bobbin material plays an important role: conductive materials such as aluminum can induce eddy-current damping, whereas non-conductive materials like plastics minimize energy loss and improve dynamic response [33].

The coil's motion is constrained by a **suspension system**, which provides both restoring stiffness and passive damping. This suspension is usually represented as a spring–damper system with parameters k and c . Together with the mass of the moving coil and bobbin m , the suspension defines the resonance characteristics of the actuator [32, 30]. The resulting lumped-parameter mechanical model is shown in Fig. 3.1.

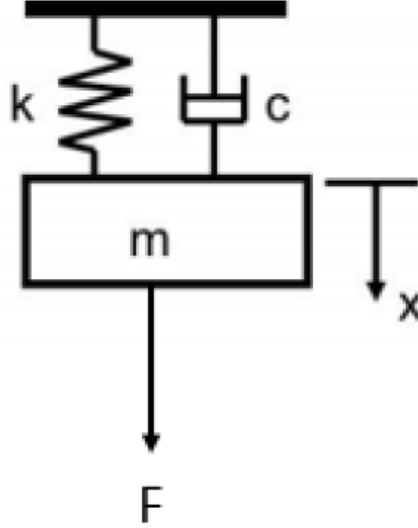


Figure 3.1 Mechanical equivalent model of the VCA represented as a mass-spring-damper system.

All components are housed in a rigid casing, which provides structural stability and ensures proper thermal dissipation. Joule heating in the coil can increase winding resistance; therefore, efficient thermal management is essential.

Since each physical component of the actuator has a direct counterpart in the mathematical description, the following section derives the governing equations. The magnet assembly defines the magnetic flux density in the air gap, the coil generates force proportional to the current, and the suspension contributes stiffness and damping. Together, these elements form a coupled electromechanical system described below.

3.1.1 Mathematical Modeling of a Voice Coil Actuator

Electrical Subsystem

Applying Kirchhoff's voltage law to the coil yields:

$$u(t) = Ri(t) + L \frac{di(t)}{dt} + e(t), \quad (3.1)$$

where $u(t)$ is the applied voltage, R the coil resistance, L the inductance, and $e(t)$ the back electromotive force (BEMF) [32, 30].

3. Analysis of the Voice Coil Actuator System

The BEMF is proportional to the velocity of the moving coil:

$$e(t) = k_b \dot{x}(t), \quad (3.2)$$

with k_b denoting the back-EMF constant [31].

3.1.2 Mechanical Subsystem

The electromagnetic force generated is proportional to the current:

$$F_{em}(t) = k_f i(t), \quad (3.3)$$

where k_f is the force constant [29, 31].

According to Newton's second law, the dynamics of the moving element are given by:

$$m\ddot{x}(t) + c\dot{x}(t) + kx(t) = F_{em}(t), \quad (3.4)$$

with m the moving mass, c the damping coefficient, and k the suspension stiffness [32, 30].

The electrical dynamics of the coil can be represented by the equivalent circuit shown in Fig. 3.2, which includes R , L , and the BEMF source.

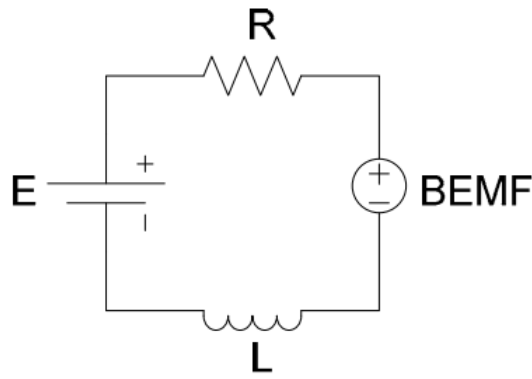


Fig. 4a

Figure 3.2 Electrical equivalent circuit of the VCA coil including resistance R , inductance L , and back-EMF source.

3.1.3 Coupled Electromechanical Model

By substituting Eq. (3.3) into Eq. (3.4), the coupled equation of motion becomes:

$$m\ddot{x}(t) + c\dot{x}(t) + kx(t) = k_f i(t). \quad (3.5)$$

Together, Eqs. (3.1)–(3.5) form the electromechanical model of the actuator [32, 30].

Taking Laplace transforms (zero initial conditions) gives:

$$U(s) = RI(s) + LsI(s) + k_b sX(s), \quad (3.6)$$

$$ms^2 X(s) + csX(s) + kX(s) = k_f I(s). \quad (3.7)$$

Eliminating $I(s)$ yields the transfer function from input voltage to displacement:

$$\frac{X(s)}{U(s)} = \frac{k_f}{mLs^3 + (mR + cL)s^2 + (cR + kL + k_b k_f)s + kR}. \quad (3.8)$$

This shows that the actuator behaves as a third-order electromechanical system, with the coupling term $k_b k_f$ representing the back-EMF effect [32].

The derived model provides the theoretical basis for analyzing actuator dynamics. In order to identify parameters and validate the model, the next section presents the experimental setup and measurement procedures used in this work.

3.2 Analysis of the Current VCA-Based System

3.3 Overview of the Current VCA System

This section provides an overview of the existing Voice Coil Actuator (VCA)-based setup. The System consists of seven main parts.

- Spring frame (Minimizing the loss of vertical motion transmitted to the node)
- Magnet Housing (fixed Magnetic field is always formed)
- Bobbin Coil (Magnetic field is formed only when current flows)
- Node (Transmitting vertical motion directly to the human body as sound and vibration)
- Node screw (fixes the node to the bobbin coil)
- Rubber frame (Suppresses vibration from the body from being transmitted to the outside world)
- Connection PCB (Take the analog signal from the AMP and apply it to the bobbin coil)

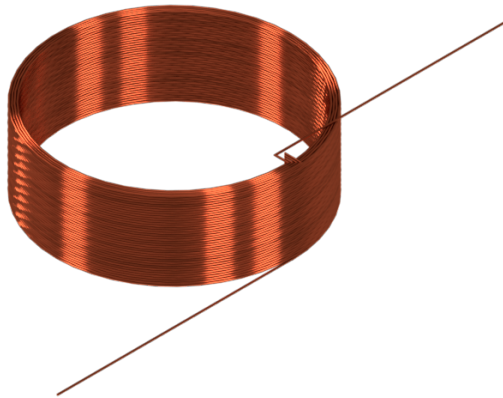


Figure 3.3 coil model

3. Analysis of the Voice Coil Actuator System

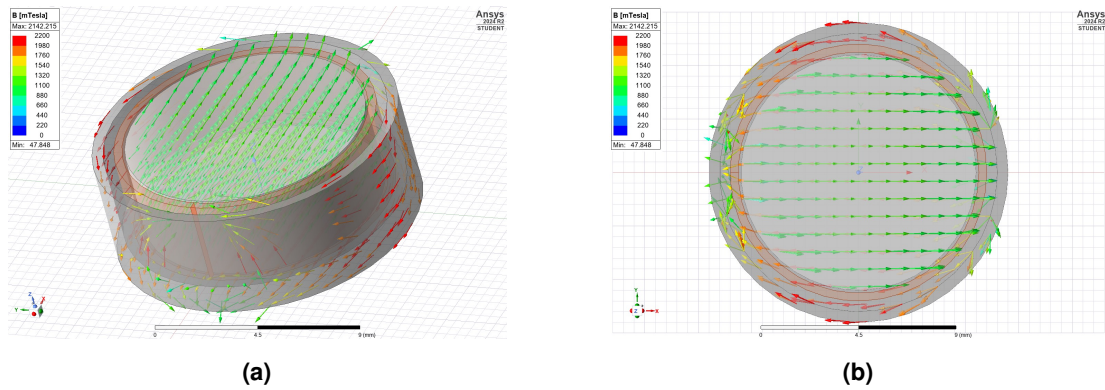


Figure 3.4 Ansys simulation of the B-field in the Magnet Housing
(a) VCA in iso view (b) VCA in top view

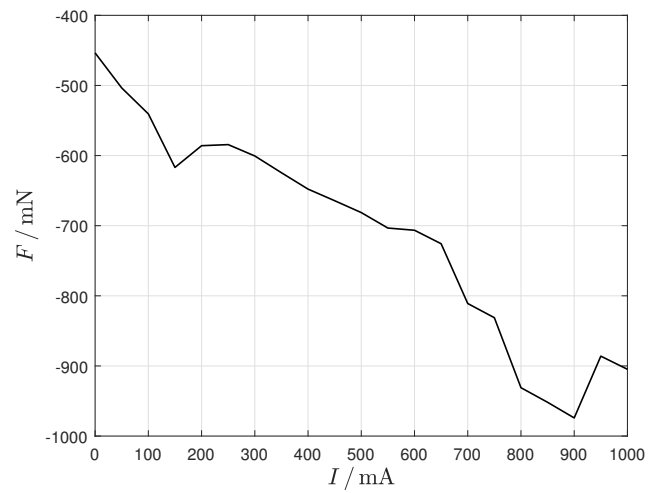


Figure 3.5 Force Plot

3.4 Dynamic Behavior: Frequency Measurement

3.4.1 Objective

3.4.2 Measurement Setup

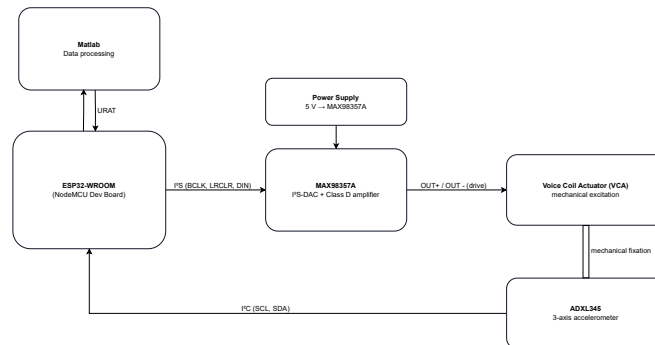


Figure 3.6 BlockdiagramMeasurementSetup

3.4.3 Results & Interpretation

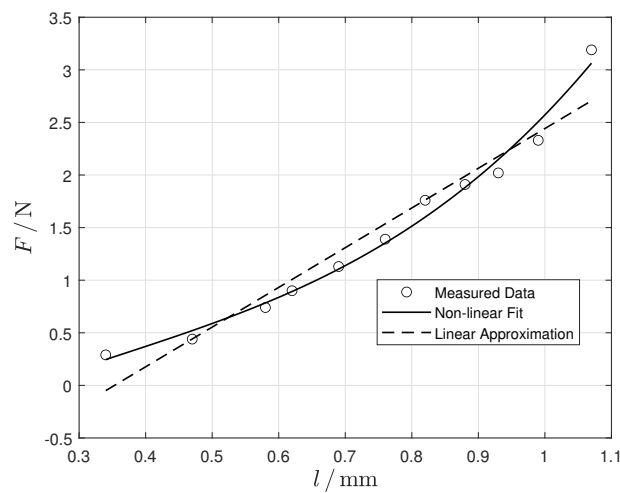


Figure 3.7 SpringTest

3. Analysis of the Voice Coil Actuator System

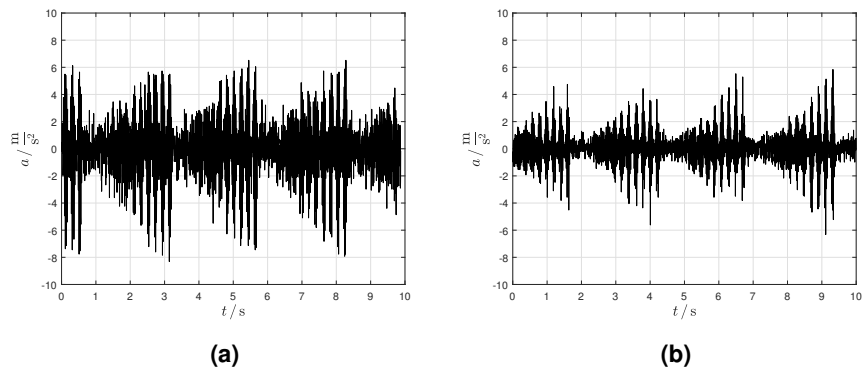


Figure 3.8 (a) With 36g Mass (b) Without Mass

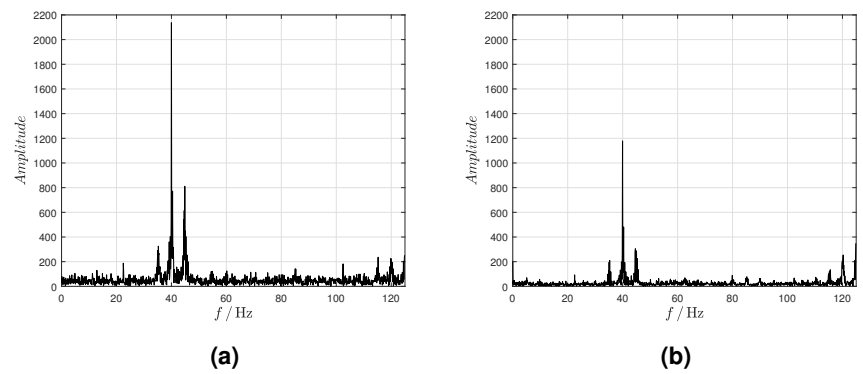


Figure 3.9 (a) With 36g Mass (b) Without Mass

3. Analysis of the Voice Coil Actuator System

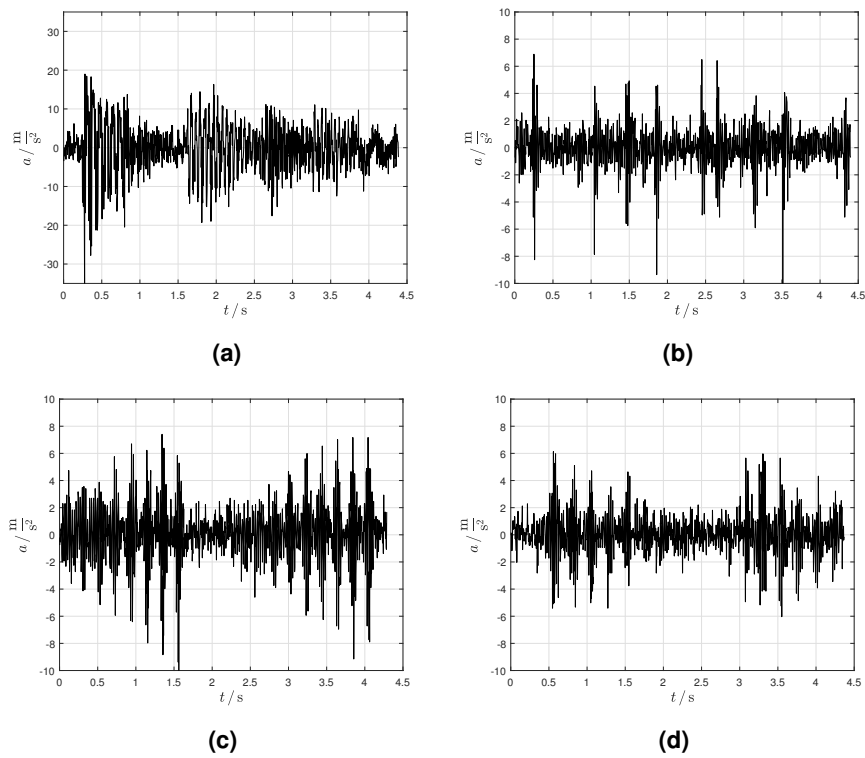


Figure 3.10 (a) Classic Mode (b) Energy Mode (c) Healing Mode (d) Recovery Mode

3. Analysis of the Voice Coil Actuator System

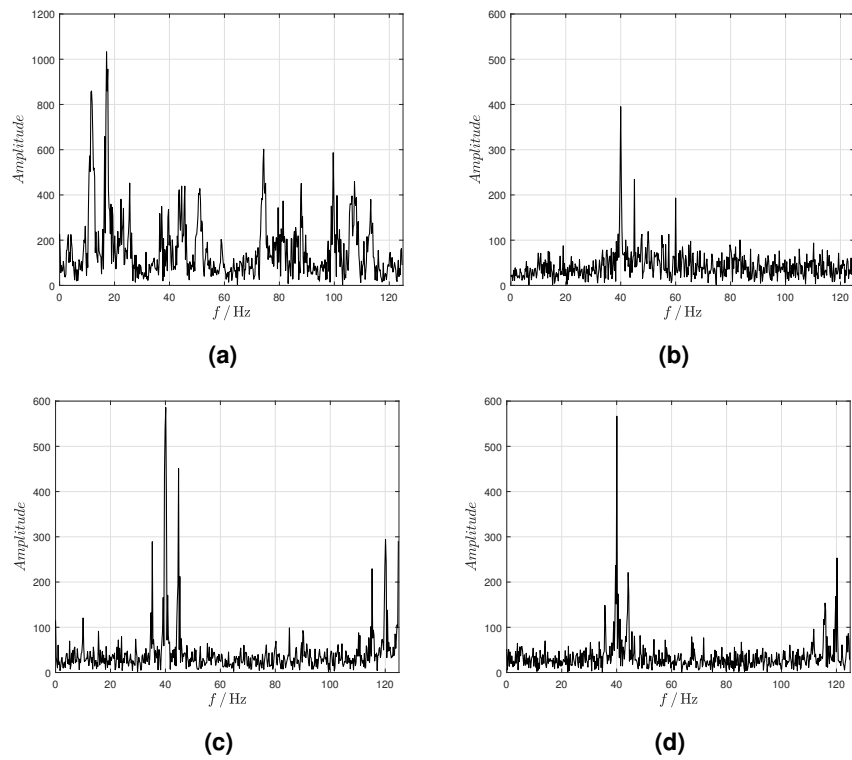


Figure 3.11 (a) Classic Mode (b) Energy Mode (c) Healing Mode (d) Recovery Mode

3.5 Limitations and Identified Challenges

4 Modify

4.1 Designanpassung

4.2 Prototyping

5 Evaluation

6 Conclusion

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Table 1 Comparison of studies on tactile low-frequency vibration in dementia management (based on Campbell et al., 2022)

Study	Type of Vibration	Duration / Frequency	Outcomes	Application Context
[19]	Vibroacoustic (40 Hz, music, physioacoustic chair)	2x week, 6 weeks	Improved SLUMS scores, increased attention	Outpatient facility
[20]	Vibroacoustic (40 Hz, daily home application)	Daily, 3 years	Stable MMSE scores over 3 years, reduced frustration	Home use
[23]	Mechanical (WBV, frequency increase from 20 to 35 Hz)	5x/week, 8 weeks	Significant EEG activation, cognitive improvement	Community centers
[17]	Mechanical (WBV, 30 Hz, 2 mm amplitude)	2x/week, 9 weeks	Improved mobility, balance, high participation rate	Day-care center
[18]	Mechanical (WBV, 30 Hz, 1–2 mm amplitude)	Multiple times/week, 12 min	Good acceptance, some reported side effects (nausea)	Nursing home